

CFD Simulation and Comprehensive Data Visualization in eRobotics Systems for Storm Damage Prevention in Forest Planning

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Abstract—CFD simulation tools typically are expert systems and demand for specific expertise; moreover, commonly provided visualizations of the simulation results are hard to analyze and interpret, especially to people not closely related to the specific area of expertise. As a recent example, the need for such simulations, as well as tools for comprehensive visualizations and analysis of the generated results became clear in 2007 when the windstorm "Kyrill" hit Europe and caused massive damages to the forests. Proper storm simulations can support forest planning in order to prevent future storm damages. The recently introduced concepts of eRobotics aim to bridge the gap between expert systems and comprehensive software applications by supporting the understanding of complex simulation results with realistic virtual environments. In this paper, we extend the eRobotics approach to the forest planning domain. Accurate CFD simulations are combined with comprehensive visualization approaches in order to act as a decision support system to optimize future forest developments under ecological and sustainable aspects. Finally, a comparison of real-world and simulated storm damage caused by the windstorm Kyrill will show the applicability and benefits of the presented system regarding damage prevention and sustainable forest development.

Keywords—computational fluid dynamics; data visualization; VR simulation systems; eRobotics;

I. INTRODUCTION

The simulation of flows of fluid substances is a major field of research in many different areas of computer science and engineering and widely applied in virtual prototyping. Apart from these technical domains, CFD simulations can also be applied in the fields of forest planning where the inclusion of ecological and sustainable aspects influence the future development of the forest. A recent example was the windstorm Kyrill that hit Germany in 2007 and caused massive damages, which have been quantified to 2.4 billion Euro (5.8 billion Euro across whole Europe). Approximately half of the costs in Germany and one third of the costs in Europe were related to damages cause to the forests. Due to the fact that especially spruces are shallow-rooted tree species that are not very resistant to strong winds, approximately 15% of the whole spruce stands in the German state of North Rhine-Westphalia were destroyed in one night, which summed up to 15.7 million solid cubic

meters of round timber [1]. The insurance industry classified Kyrill as the most expensive natural disaster in Europe. In order to prevent such enormous ecological and economic damages in the future, storm simulations are an essential task in forest planning domains. Regarding the growing number of occurring windstorms and resulting damage over the last decades, simulation technology and tools for comprehensive analysis and interpretation of the generated results are becoming increasingly important, even in domains like forest planning, which have not been a typical application area in the past.

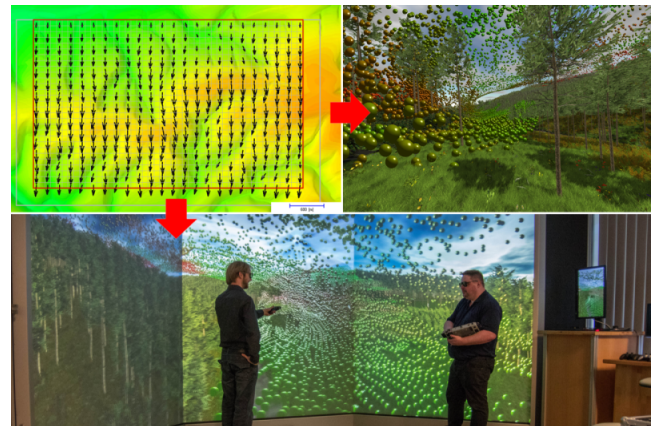


Figure 1. Comprehensive visualization and analysis of the results of a storm simulation in an interactive multi-screen 3D projection.

In this contribution, we introduce a framework that focuses on the convenient integration of accurate CFD simulations in forestry domains as well as the intuitive visualization of the generated storm simulations for comprehensive analysis in realistic virtual environments as illustrated in Figure 1. In Section II, related work regarding the eRobotics approach, CFD simulations and data visualization are introduced. Section III details the integration into an eRobotics-capable system and compares multiple interactive data visualization approaches regarding its appliance in forest planning domains. Section IV shows the practical application of the presented approaches and presents a case

study that compares simulated and actual storm damages caused by the windstorm Kyrill in the Schmallingberg area in Germany in order to demonstrate the simulation accuracy, as well as the benefits arising from a comprehensive data visualization. Finally, Section V concludes this work and suggests future work.

II. RELATED WORK

In recent papers, we introduced concepts and application areas of eRobotics [2][3]. One of the goals of eRobotics is to combine realistic computer graphics with complex simulations in order to improve the comprehensibility and accessibility of complex datasets or simulation results [4]. The broad range of available data is not only used to improve the accuracy of simulation tasks - like fluid dynamics in this case - but also for the generation of a realistic virtual mapping of the actual environment, predestinated for intuitive data visualization of the performed storm simulations. Finally, eRobotics applications aim to apply the GPU for simulation purposes in order to improve accuracy and performance. Data visualization approaches are suitable examples for this application area.

Currently, a broad range of CFD simulators exists for different domains. Traditionally, CFD calculations are closely related to high-performance computing due to their computational complexity and, consequently, they are separated from real-time vector field visualization approaches. Even though GPU-based CFD simulations exist that similarly simulate and visualize the fluid flow in real-time [5] [6], these approaches are mainly intended for game engines due to the fact that they use a simple but rapid Navier-Stokes solver for incompressible fluids geared towards visual quality and performance instead of physical realism. Regarding the desired storm simulation, the CFD software toolkit WindStation is an ideal choice [7]. It focuses on the accurate calculation of airflow over complex topography and uses the CFD solver Canyon, which is a generalized coordinate system solver based on a control-volume approach [8]. Canyon solves the full 3D Navier-Stokes for mass conservation, momentum conservation, energy conservation and turbulence quantities using the $k-\epsilon$ model[9]. Terrain roughness is also taken into account through the ground shear stress modeling as well as temperature imposition or heat flux imposition. WindStation also features simple visualizations as shown in the top-left image of Figure 1.

As shown in Figure 2 and stated by Weiskopf [10], visualization approaches can roughly be categorized into direct, texture-based and geometry-based visualizations. Popular examples for direct visualizations are hedgehogs, oriented lines as well as glyph-based approaches, which aim to show local features of the vector field. Texture-based approaches process the vector field in order to display two- or three-dimensional output textures. Popular approaches are based on line integral convolution (LIC) introduced by Cabral

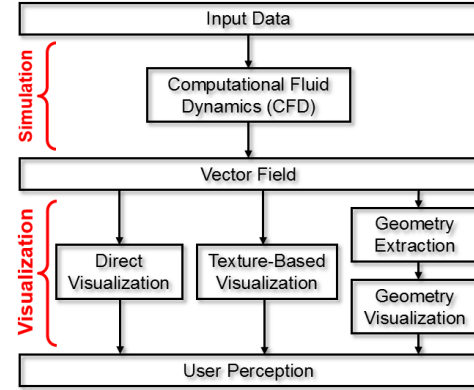


Figure 2. Illustration of the common simulation and visualization process for flow simulations.

and Leedom [11]. The key idea is to blur an input texture along field lines from a given flow vector field in order to properly visualize the specific characteristics of a fluid flow. Oriented Line Integral Convolution (OLIC) further integrates the information of the flow direction into the visualization [12]. Recent approaches also apply GPU-based LIC-techniques in order to visualize multivariate data [13]. Finally, geometry-based visualization approaches aim to understand the specific flow properties of the vector field by moving objects through it. These approaches are known under the term particle tracing [14].

III. WIND SIMULATION AND VISUALIZATION FOR FOREST PLANNING DOMAINS

The approaches presented in this paper are part of the Virtual Forest Project [15], which goal is to provide a comprehensive multi purpose software tool for forestry-related domains. As illustrated in Figure 2, the implemented wind simulation is separated into a simulation and visualization part. There are two reasons for this: first, this approach grants maximum flexibility. The vector field generator can easily be exchanged without the need to modify the visualization module and vice versa. Second, an accurate flow simulation is computationally expensive and usually carried out on the CPU while visualization approaches can efficiently be implemented on the GPU and run in real-time. Even if some real-time GPU-based CFD simulations exist, they only use simplified simulation models and only hold for incompressible fluids due to performance reasons. Highly accurate, high-resolution vector fields can be calculated in an offline process but visualized in a real-time environment, giving the user the possibility to interactively explore the generated flow field in intuitive ways. The implementation into the eRobotics system is illustrated in Figure 3.

The WindStation toolkit is used to calculate the wind vector field. Required information like the topography or ground roughness are exported from the semantic database,

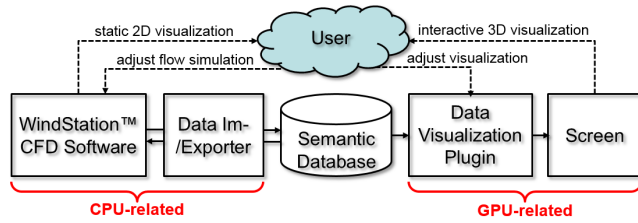


Figure 3. Illustration of the implemented flow simulation and visualization pipeline.

which contains all gathered data related to the Virtual Forest Project. The user can conveniently set up the wind simulation parameters and have a first look at the results before the data is imported to the eRobotics system for further analysis with advanced visualization approaches that can be displayed in highly realistic virtual model of the actual environment. In the following sub-sections, approaches related to the three visualization categories will be detailed and compared regarding its application in forest planning domains. For comparison reasons, all approaches are applied to the same vector field.

A. Direct Visualization

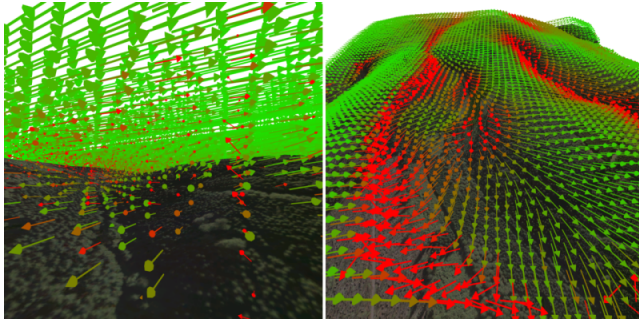


Figure 4. Glyph-based vector field visualizations. Left: full vector field visualization. Right: 2.5D visualization.

Direct visualizations show magnitude and direction of the vector field by using glyphs. Additional data - like pressure, turbulence, temperature, etc. - can be visualized by color coding. Figure 4 shows screenshots of a direct visualization of the wind vector field using arrow glyphs. In the left image, the arrows point in wind direction, the wind speed is visualized by the size of the arrows and the arrows colors represent the local turbulence value (green indicates low turbulence values, red indicates a high turbulence value). While this approach can easily be implemented and achieves clear visualization results in 3D, the figure also illustrates some disadvantages. Firstly, the arrow plot visualization is not suitable for dense or large vector fields due to inherent occlusion effects. Secondly, the arrow plots get unclear if too much information is encoded into their visualization. The

high wind speed in higher altitudes generates large arrows that cover large areas of the image. Relevant information in lower altitudes, like turbulences close to the ground with relative low wind speed, are hard to notice in comparison. Finally, the visualization results are poor if the magnitude of the velocity changes rapidly. A possible solution to address the named limitations is the reduction of either the amount of visualized information or the amount of arrow glyphs. Specific areas of interest can be defined or a 2D slice of arrows can be visualized for a given height above the ground. The right image of Figure 4 shows a corresponding partial vector field visualization at 10 meters above ground level. The arrows glyphs point in wind direction, the turbulence is encoded into the color and the wind speed is neglected. This visualization method is commonly known as 2.5D visualization, as it projects a single slice of the vector field into the 3D geometry. Relevant Information can more clearly be recognized compared to the full 3D vector field visualization.

B. Texture-Based Visualization

Multiple texture-based visualization approaches have been implemented based on line integral convolution (LIC). For the practical application and proper interpretation of the results, the generated textures can directly be projected onto the scene geometry as illustrated in the left image of Figure 5, which shows the results of a standard LIC approach. The textures were generated for the same height 10 meters above ground level as the 2.5D visualization shown in Figure 4. Important information like local turbulences are clearly visible at the corresponding areas, which allows for an intuitive interpretation of the flow simulation results. The right image shows the results of an OLIC approach, which also depicts the wind direction.

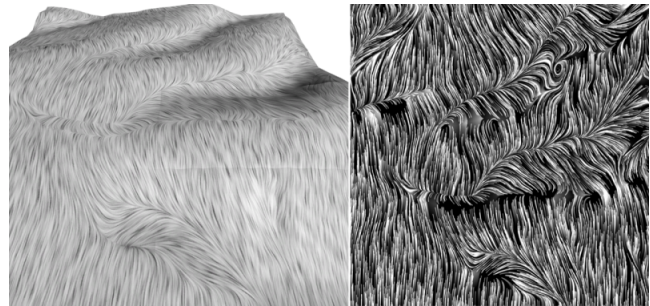


Figure 5. Left: traditional LIC visualization mapped onto the 3D scene. Right: OLIC visualization of the same region that also emphasizes the wind direction.

In order to support the understanding of the generated LIC textures hybrid approaches are suitable, which also include additional flow properties or environmental information. The left image of Figure 6 shows such an implementation that blends the OLIC texture with the matching ortho photos

of the specific area in order to put the simulation results into relation with the actual environment. Regarding the application in forest planning domains, the local turbulence value is an important information. Strong turbulences near the ground can lead to damages to the forest; hence, it must not be neglected in the visualization. The right image shows a visualization that is provided in addition to the left one. This visualization color-codes the turbulence values into the red color channel. In order to prevent a confusing, overloaded visualization, the LIC texture is not included directly; instead, the convolution is applied to the ortho photo and smears the image in wind direction in order to provide information about the wind direction and speed. By providing both visualizations, the forest planner has all the important information precisely and comprehensively in view at all times.



Figure 6. Left: OLIC-visualization blended over the surface's ortho photo. Right: the ortho photo is used as LIC input texture; additionally, the local turbulence value is added to the red color channel.

C. Geometry-Based Visualization

The approaches presented so far only provide static information about the flow. It becomes clear that the visualized information has to be reduced to a single height layer or just a few flow properties in order to keep these visualizations clear and understandable. By using a dynamic approach that moves particles through the vector field, geometry-based visualizations provide the possibility to overcome these limitations and similarly provide the most natural kind of visualizing a flow. Particle tracing requires the solution to an ordinary differential equation (ODE) in order to computing the trajectory of massless particles in a flow field over time with an initial condition of $x_p(0) = x_0$ for a numerical solution:

$$\frac{\partial x_p}{\partial t} = \tilde{v}(x_p(t), t) \quad (1)$$

, where $x_p(t)$ is the time-dependent particle position, $\frac{\partial x_p}{\partial t}$ is the tangent to the particle trajectory and $\tilde{v}$ the approximation of the vector field v . CFD calculations generate v with a discrete resolution, hence, an appropriate interpolation need to be used in order to reconstruct particle velocities along

their characteristic paths. Fixed step size integration schemes like the classical Euler (see Equation 2) or Runge-Kutta (see Equation 3) are common for visualization reasons.

$$x_{t+1} = x_t + \Delta t \cdot f'(t, x_j) \quad (2)$$

$$x_{t+1} = x_t + \frac{\Delta t}{6} \cdot (k_1 + 2k_2 + 2k_3 + k_4) \quad (3)$$

where

$$k_1 = f'(t, x_j)$$

$$k_2 = f'(t + \frac{\Delta t}{2}, x_j + \frac{\Delta t}{2}k_1)$$

$$k_3 = f'(t + \frac{\Delta t}{2}, x_j + \frac{\Delta t}{2}k_2)$$

$$k_4 = f'(t + \Delta t, x_j + \Delta tk_3)$$

The Euler integration scheme provides a good compromise between precision and performance and was integrated in the particle tracer. With the current implementation and a common desktop PC equipped with a modern GPU, hundreds of thousands particles can be traced and visualized in real-time. Figure 7 shows animation sequences of multiple, currently implemented real-time particle tracing visualizations. Local turbulences can easily be recognized by the particles movement while the turbulence value is encoded into the green and red color channel. Finally, illuminated streamlines can be applied to clearly trace the wind direction in the 3D environment.

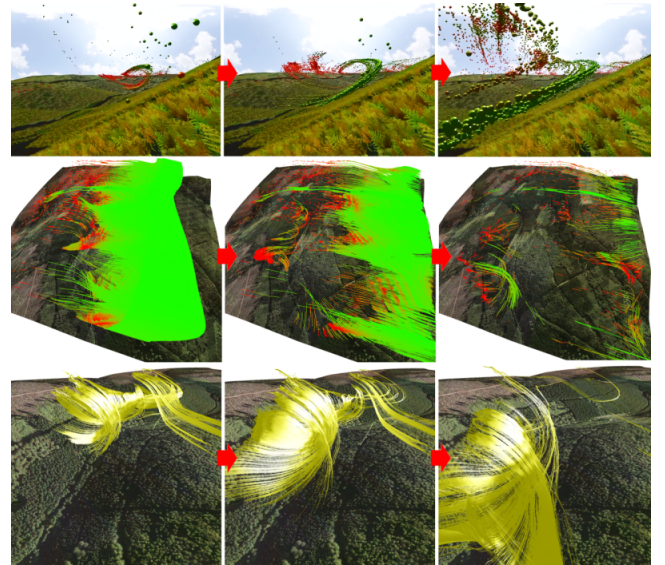


Figure 7. Top row: movement of sphere sprites through the flow field. Middle row: line segments (linelets). Bottom row: illuminated streamlines.

IV. APPLICATIONS AND RESULTS

For evaluation reasons, a CFD simulation for an actual area in Schmallenberg, Germany was calculated so that it matches the actual parameters of the windstorm Kyrill. The top image of Figure 8 shows an aerial photo of that region where the storm damaged the forest in a specific area.

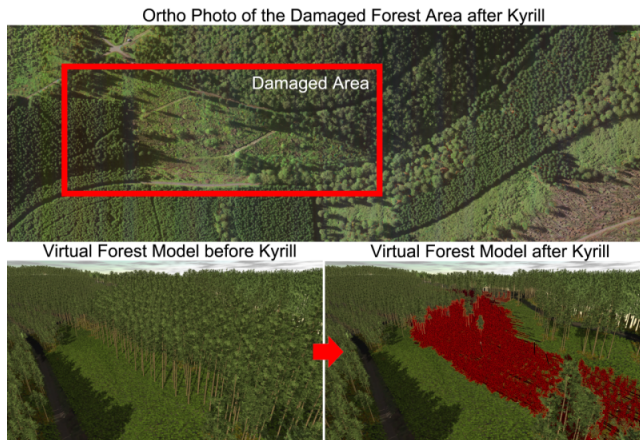


Figure 8. Top: ortho photo of the damaged stand in the Schmallenberg region in Germany. Bottom row: virtual model of the stand before and after the windstorm.

The bottom images show the virtual 3D model before and after the storm with the damaged area marked in red. As earlier stated, eRobotics systems aim to visualize complex simulation results in attractive virtual environments. As part of the Virtual Forest Project, relevant data like the surface topography and terrain roughness for an accurate wind simulation as well as a close-to-reality virtual mapping of the real forest environment are already available [2]. Figure 9 shows an animation sequence of approximately 100.000 particles, which are traced through the vector field in real-time. This visualization reveals the reason for the massive damage to the specific area. A massive wind front was dammed at the preliminary ridge that subsequently breaks at the shown location, causing heavy turbulences, which lead to heavy damage in the stand. This example illustrates how essential proper flow simulations and visualizations are in forest planning domains; as a consequence, windbreaks can be built or more robust, deep-rooted tree species like beeches or oaks can be planted in high-risk regions.

CFD simulation results become easily understandably and that possible damages to the forest can be analyzed and identified more easily when the user can actually navigate through the actual 3D forest environment. As already illustrated in Figure 1, the presented visualization approaches are integrated into these realistic 3D models and drastically improve the understanding of complex simulation data compared to common, simple 2D visualizations. The user can conveniently alter the wind parameters in the CFD simulation and create a new vector field, which is loaded

into the simulation system. In the 3D simulation, he can switch through the presented visualization approaches and locally release particles in the 3D simulation in order to further analyze a specific region under different conditions.

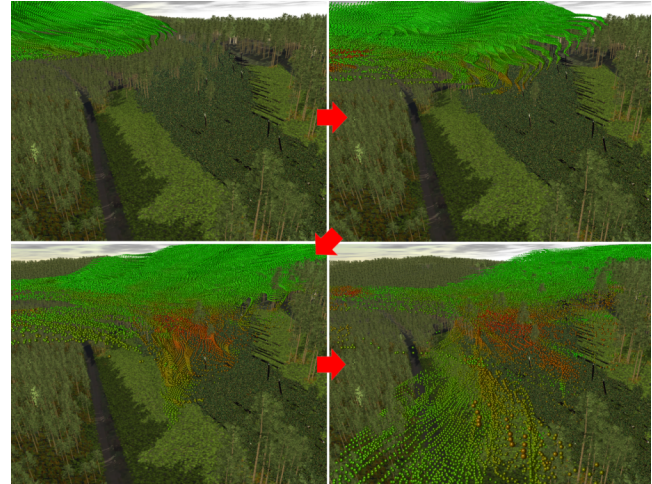


Figure 9. The interactive particle tracer clearly reveals the reason for the damage caused to the corresponding stand.

V. CONCLUSION AND FUTURE WORK

In this paper, we presented a software framework for CFD simulations and visualizations for forest planning domains. The presented tool chain allows for convenient setup of matching wind parameters for the CFD simulation while the eRobotics system clearly improves the understanding of the generated results, which are presented in an attractive virtual mapping of the real environment. Multiple real-time visualization approaches have been presented and compared, which are suitable for the data analysis in the desired domains. The presented system can be used as a decision support system for the forest planner, in order to identify high-risk regions and to prevent future storm damage. A comparison of the simulation results and an actual windfall region demonstrated the accuracy and practicability of the presented system.

Our future developments will focus on the prediction of storm damage directly in the simulation based on the calculated wind vector field in order to provide a holistic but comprehensive software tool for forest planning domains, which further expands the range of eRobotics applications.

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